St. Philomena's College (Autonomous), Mysore

PG Department of Studies and Research in Chemistry

Question Bank (Revised Curriculum 2018 onwards)

First Year - FourthSemester (2018 -20 Batch)

Course Title (Paper Title):

Molecular symmetry, Statistical thermodynamics, Polymers and Radiation Chemistry

QP Code:55302

Unit	SI. No	Questions	Marks
1	1	Identify the symmetry elements present in N_2H_2 (trans structure)	02
1	2	Identify the symmetry elements present in Naphthalene	02
1	3	Identify the symmetry elements present in $POCl_3$	02
1	4	Identify the point group to which the following molecules belong PF_5, PCl_3	02
1	5	Write down all the symmetry elements of methane	02
1	6	Among the following, why does only XeF_4 contain a 4-fold principal rotation axis: CF_4, SF_4, $[BF_4]$- and XeF_4 ?	02
1	7	N_2O_4 is a planar. Show that it possess three planes of symmetry	02
1	8	Why does CO_2 possess an inversion centre, but NO_2 does not?	02
1	9	Show that the planes of symmetry in benzene are one σh, three σv and three σd.	02
1	10	How many planes of symmetry do (a) $F_2C=O$, (b) $ClFC=O$ and c) $[HCO_2]$- possess?	02
1	11	What symmetry operators are lost in going from NH_3 to NH_2Cl?	02
1	12	Compare the symmetry operators possessed by NH_3, NH_2Cl, $NHCl_2$ and NCl_3.	02
1	13	Draw a diagram to show the symmetry operators of $NClF_2$.	02
1	14	Show that BF_3 and $F_2C=O$ have the following symmetry elements in common E, two mirror planes, one C_2.	02
1	15	How do the symmetry elements of ClF_3 and BF_3 differ?	02
1	16	What symmetry elements do BCl_3 and PCl_3 (a) have in common and (b) not	02

		have in common?	
1	17	Draw the structure of SO_2 and identify its symmetry, properties.	02
1	18	Which of the following molecules contains a C_4 axis and a σ_h plane: CCl_4; $[ICl_4]^-$;sulphate; SiF_4; XeF_4 ?	02
1	19	How many mirror planes do each of the following molecules contain: a)SF_4; b) H_2S; c)SF_6; d) SOF_4; e) SO_2; f) SO_3 ?	02
1	20	Which of the following species contain inversion centers? a) BF3; b) SiF4; c) XeF4; d) PF5; e) XeF_5; f) SF_6; g)C_2F_4; f) SF_6	02
1	21	Write the signs of 8 octants in a cube of the x,y,z molecular planes	02
1	22	List the groups which are the subgroups of D_{6h} and C_{3v}	02
1	23	List the groups which are the subgroups of C_{2v} and T_d	02
1	24	List the groups which are the subgroups of C_{2v} and O_h	02
1	25	List the groups which are the subgroups C_{4v} and C_{2h}	02
1	26	List the groups which are the subgroups D_{2h} and D_{2d}	02
1	27	List the groups which are the subgroups $C_i$and C_s	02
1	28	Identify the point group to which the following molecules belong PCl_3F_2, SF_4BrCl	02
1	29	Identify the point group to which the following molecules belong Cis-butadiene, transbutadiene,	02
1	30	Identify the point group to which the following molecules belong a) hexatriene b) cis-dichloroethene,	02
1	31	Identify the point group to which the following molecules belong boat form cyclohexane, NH_2F	02
1	32	Identify the point group to which the following molecules belong carbonate, sulphate ion	02
1	33	Identify the point group to which the following molecules belong monoclorobenzene, Ortho-dichlorobenzene	02
1	34	Identify the point group to which the following molecules belong Mono chloroacetylene, dichloroacetylene	02

1	35	Identify the point group to which the following molecules belong hydrogen peroxide, formaldehyde	02
1	36	Identify the point group to which the following molecules belong CH_3F, CH_2F_2	02
1	37	To what point group does $POCl_3$ belong?	02
1	38	Assign the point groups to ammonium and ammonia molecular systems	02
1	39	To which point group does NF_3 belong?	02
1	40	The point group of $[AuCl_2]^-$ is D∞h What shape is this ion?	02
1	41	Determine the point group of SF_5Cl.	02
1	42	Assign a point group to each member in the series (a) CCl_4, (b) CCl_3F, (c) CCl_2F2, (d) $CClF_3$ and (e) CF_4.	02
1	43	Deduce the point group of SF4. Is SOF_4 in the same point group?	02
1	44	Which of the following point groups possesses the highest number of symmetry elements: (a) O_h; (b) T_d; (c) I_h?	02
1	45	$[PtCl_4]^{2-}$ has a centre of symmetry but $[CoCl_4]^{2-}$ does not. One is square planar and the other is tetrahedral which is which?	02
1	46	CS_2 and HCN are both linear. Explain why CS_2 possesses a centre of symmetry whereas HCN does not.	02
1	47	Show that operations inversion and rotation commute	02
1	48	Show the abelian nature of C_s point group	02
1	49	Show the abelian nature of C_i point group	02
1	50	Show the abelian nature of S_4 point group	02
1	51	Show the abelian nature of C_4 point group	02
1	52	$[PtCl_4]^{2-}$ is square planar to what rotation operation is equivalent?	02
1	53	Explain what is meant by an∞-fold axis of rotation.	02
1	54	Write the group multiplication table for C_2	02
1	55	The following are sets of numbers. Which of them from a group? The mode of combination is given in parenthesis a) 1,-1, 0 (addition) b) 2,-2,0 (addition)	02

1	56	The following are sets of numbers. Which of them from a group? The mode of combination is given in parenthesis a) i,-1,+1, -i (multiplication) b) i,-1,+1, -i (addition)	02
1	57	The numbers ±1 and ±i form a group under multiplication	02
1	58	What do you mean by reducible and irreducible representation	02
1	59	Why do CO_2 and SO_2 have a different number of degrees of vibrational freedom?	02
1	60	How many degrees of vibrational freedom do each of the following possess: $SiCl_4$, BrF_3, $POCl_3$?	02
1	61	How many IR active vibrational modes does CS_2 possess, and why? Hint: CS_2 is isostructural with CO_2.	02
1	62	Determine the number of degrees of vibrational freedom for each of the following: (a) SO_2; (b) SiH_4; (c) HCN; (d) H_2O; (e) BF_3.	02
1	63	How many normal modes of vibration are IR active for (a) H_2O, (b) SiF_4, (c) PCl_3, (d) $AlCl_3$, (e) CS_2 and (f) HCN?	02
2	64	Define micro and macrostates of a system	02
2	65	Define thermodynamic probability of a system.	02
2	66	Write Boltzman-plank equation and explain the terms involved in it.	02
2	67	Define partition function and write the expression for partition function	02
2	68	Write an expression for translational partition function and explain the terms involved in it	02
2	69	Write an expression for rotational partition function and explain the terms invoved in it	02
2	70	Calculate the thermodynamic parameter entropy by using the concept of partition function	02
2	71	Calculate the thermodynamic parameter enthalphy by using the concept of partition function	02
2	72	Calculate the thermodynamic parameter Helmohltz free energy by using the concept of partition function	02
2	73	Calculate the thermodynamic parameter gibbs free energy by using the	02

		concept of partition function	
2	74	Calculate the thermodynamic parameter pressure by using the concept of partition function	02
2	75	Define Gibb's Phase rule and explain the terms involved in it	02
2	76	Write the mathematical form of gibbs phase rule and explain the terms involved in it.	02
2	77	Define phase of a sytem with example.	02
2	78	Define component of a sysytem with example	02
2	79	What is degree of freedom	02
2	80	What is the effect of pressure on energetics of cell reaction	02
2	81	What is the effect of temperature and concentration on energetics of cell reaction	02
2	82	Write any two applications of phase rule.	02
2	83	What are batteries and write its classification.	02
2	84	What are lithium batteries give example.	02
3	85	What is radiation chemistry?	02
3	86	How radiation interact with matters?	02
3	87	Write one similarity and difference of X-rays and gamma rays.	02
3	88	Compare the difference between X-rays and gamma-rays.	02
3	89	Write the four principal mode of interaction of gamma radiation with matters	02
3	90	What is Photoelectric effect? Write the equation for representing electron ejecting energy.	02
3	91	Give the difference between θ and φ in Compton effect.	02
3	92	Define (i) rad (ii) Gray	02
3	93	Write the SI unit of dose in radiation energy	02
3	94	What is RBE and Rem in radiation chemistry? Hence, Convert 1 sievert into rem.	02

3	95	What is G value? Write the formula to calculate the G value and explain it's terms.	02
3	96	Define G value, How to denote G value for decomposition of $CaCO_3$	02
3	97	Briefly explain the difference between radiation dosimetry and chemical dosimetry.	02
3	98	Write the condition to satisfy by chemical dosimetry	02
3	99	How to prepare Fricke dosimeter solution?	02
3	100	Write the equation involved in Fricke dosimetry.	02
3	101	What is Fricke dosimeter? Hence write the accepted G value of Fe^{+3}i'eG(Fe^{+3}).	02
3	102	Write the value of pH when yield of H_2S is maximum and hence write the G (Cysteine) in radiolysis of cysteine amino acid.	02
3	103	What happens to aqueous solution of cysteine when it irradiated for longer duration during radiolysis?	02
3	104	Aromatic compounds are more stable than towards radiation than aliphatic compound Give reason with example.	02
3	105	Write the reaction involved in radiolysis of biphenyl	02
3	106	How radioisotope used in as tracers	02
3	107	Mention the points on which radioisotope used as tracer. Mention any four field where radioisotopes used as tracer	02
3	108	What is the principle for determination of solubility of springly soluble salt using radioisotope tracer	02
3	109	With equation of decay constant in carbon -14 dating and explain the term	02
3	110	What are the two types of radioisotopes used in medical field?	02
3	111	What is the use of Na-24 and I-131 radioisotope in detection purpose?	02
3	112	What is specific activity, Also write principle of sparingly soluble salt determination using radioisotope.	02
3	113	What are monomer? Give an example.	02

3	114	Give two example of repeating unit in polymer?	02
3	115	Define degree of polymerization? Give an example any polymer.	02
3	116	Make a difference between linear and network polymer?	02
3	117	Distinguish between branched and network polymer with an example	02
3	118	What is polymerization?	02
3	119	What is polydispersion?	02
3	120	How to determined molecular weight of any polymer.	02
3	121	What is viscosity of average molecular weight of polymer?	02
3	122	Name any two methods used for determination of molecular weight of polymer.	02
3	123	Write the principle of electrophoresis	02
3	124	What is electrophoresis? Hence, write principle of zone electrophoresis	02
3	125	Distinguish between free electrophoresis and gel electrophoresis	02
3	126	What is isoelectric point of protein?	02
3	127	Define (i) Isoelectric point (ii) Electro-osmosis.	02
3	128	What are micelles?	02
3	129	Define the term (i) Micelles (ii) Diffusion of solution	02
3	130	What are the applications of diffusion of solute in mechanism of respiratory exchange?	02
3	131	What is salting in of protein? Write its one application.	02
3	132	What is salting out of protein? Write its one application.	02
3	133	Define (i) Osmotic behavior of cells (ii) osmo regulation	02
3	134	Write any two application of osmotic behavior of cell.	02
3	135	Write any two application of osmo-regulation.	02
1	136	Write the structure and identify the symmetry elements present in the following cases and assign the point groups and pictorially represent all the elements of symmetry Pyridine , Ethane staggered form, Cis-2-butene,Trans-	05

		2- butene	
1	137	Write the structure and identify the symmetry elements present in the following cases and assign the point groups and pictorially represent all the elements of symmetry Transdichloroethylene, Cyclopentadienyl ion, Carbon dioxide, o-dichlorobenzene	05
1	138	Write the structure and identify the symmetry elements present in the following cases and assign the point groups and pictorially represent all the elements of symmetry p-dichlorobenzene, Eclipsedferrocene, Staggered ferrocene, HCl	05
1	139	Write the structure and identify the symmetry elements present in the following cases and assign the point groups and pictorially represent all the elements of symmetry CO, Benzene, Boric acid,Ammonia	05
1	140	Write the structure and identify the symmetry elements present in the following cases and assign the point groups and pictorially represent all the elements of symmetry Ethylene, Triphenyl arsine	05
1	141	Draw the structures of each of the following species and confirm that each possescentre of symmetry: CS_2, $[PF_6]$-, XeF_4, I_2, $[ICl_2]$-	05
1	142	How do the rotation axes and planes of symmetry in cis- and trans-N_2F_2 differ?	05
1	143	How do the rotation axes and planes of symmetry in Z- and E CFH=CFH differ?	05
1	144	By drawing appropriate diagrams, illustrate the fact that BF_3 possesses a 3-fold axis, three 2-fold axes, and four planes of symmetry. Give appropriate labels to these symmetry elements.	05
1	145	Deduce which symmetry elements are lost on going from (a) BF_3 to $BClF_2$ and (b) $BClF_2$ to BBrClF. (c) Which symmetry element (apart from E) is common to all three molecules?	05
1	146	Which of the following molecules or ions contain (a) a C3 axis but no σh plane, and (b) a C3 axis and a σh plane: NH_3; SO_3; PBr_3; $AlCl_3$; sulphate; nitrate ?	05
1	147	S.T that the symmetry group C_4 and the number group 1, i, -i and -1 are isomorphic	05

1	148	Define the operation improper rotation S_n. Show that the two operations viz., rotation and reflection implied in S_n are commutative	05
1	149	Show the abelian nature of C_{2v} point group	05
1	150	Show the abelian nature of C3v point group	05
1	151	Show the abelian nature of C_{2h} point group	05
1	152	Using appropriate figures show that there are no C_4 axis and σh symmetry plane in methane, but there is an S_4 axis	05
1	153	Explain why BF_3 possesses an S_3 axis, but NF_3 does not.	05
1	154	C_2H_6 in a staggered conformation possesses an S_6 axis. Show that this axis lies along the C–C bond.	05
1	155	The point group of BrF_3 is C_{2v}. Draw the structure of BrF_3 and compare your answer with the predictions of VSEPR theory.	05
1	156	Confirm that the symmetry operation of (a) inversion is equivalent to an S_2 improper rotation, and (b) reflection through a plane is equivalent to an S_1 improper rotation.	05
1	157	C_2 operations w.r.t three mutually perpendicular axes commute and the product of the two is equal to the third	05
1	158	Show that C_{2z}, C_{2x}, C_{2y} commute and the product of any two is equivalent to the third	05
1	159	Prove that The readings 1,2,3,…..12 on the dial of a clock form a group under multiplication	05
1	160	Draw the structure of $[XeF_5]^-$, on the diagram mark the C_5 axis. The molecule contain five C_2 axes. Where are these axes?	05
1	161	Determine the point group of trans-N_2F_2 using the flowchart.	05
1	162	Determine the point group of PF_5 using the flowchart.	05
1	163	Show that BF_3 belongs to the D_{3h} point group using flow chart	05
1	164	Show that OF_2 belongs to the C_{2v} point group using flow chart	05
1	165	By the flowchart, show that $CHCl_3$ possesses C_{3v} symmetry, but that CCl_4 belongs to the T_d point group.	05

1	166	Write the group multiplication table for water molecule	05
1	167	Write the group multiplication table for ammonia molecule	05
1	168	Construct the matrices for symmetry operations in the C_{2v} point group using H-1s orbitals and O-2pz orbital of water molecule (assume the z-axis to be C_2 axis). Write down the characters for this reducible representation and decompose in to its irreducible representation	05
1	169	Give the GOT and its consequences for C_{2v} character table	05
1	170	Give the GOT and its consequences for C_{3v} character table	05
1	171	Give the GOT and its consequences for C_{2h} character table	05
1	172	Find out the symmetries of the fundamentals of sulphate and carbonate.Comment on their infrared and Raman activities	05
1	173	Prove that in a Centro symmetric molecule (the symmetry element i is present) no IR active normal mode can also Raman active and vice versa	05
1	174	Verify the symmetry designations of the normal co-ordinates of water molecule	05
1	175	Find the symmetry co-ordinates which can be used to construct the normal coordinates of the trans N_2F_2 molecule.	05
1	176	Using VSEPR theory to help you, draw the structures of CF_4, XeF_4 and SF_4. Assign a point group to each molecule. Show that the number of degrees of vibrational freedom is independent of the molecular symmetry.	05
1	177	The IR spectrum of $SnCl_2$ exhibits absorptions at 352, 334 and 120 cm^{-1} what shape do these data suggest for the molecule and is this result consistent with VSEPR theory?	05
1	178	The vibrational modes of XeF_2 are at 555, 515, 213 cm^{-1} but only two are IR active. Explain why this is consistent with XeF_2 having a linear structure.	05
1	179	The IR spectrum of SF2 has absorptions at 838, 813 and 357 cm^{-1}. Explain why these data are consistent with SF_2 belonging to the C_{2v} rather than $D_{\infty h}$ point group.	05
1	180	To what point group does F_2O belong? Explain why the vibrational modes at 928, 831, and 461 cm^{-1} are all IR active.	05
1	181	The IR spectrum of BrF_3 in an argon matrix shows six absorptions Explain	05

		why this observation confirms that BrF_3 cannot have C_{3v} symmetry.	
1	182	The IR spectrum of BF_3 shows absorptions at 480, 691 and 1449 cm-1. Use these data to decide whether BF_3 has C_{3v} or D_{3h} symmetry.	05
1	183	The IR spectrum of gaseous ZrI_4 shows absorptions at 55 and 254 cm^{-1} Explain why this observation is consistent with molecules of ZrI_4 having T_d symmetry.	05
1	184	The $[PdCl_4]^{2-}$ ion gives rise to three absorption in its IR spectrum (150, 321, 161 cm^{-1}) Rationalize why this provides evidence for a D_{4h} rather than T_d structure.	05
2	185	Derive Boltzman-p;ank equation	05
2	186	Derive a relation between thermodynamic probability and entropy.	05
2	187	Discuss the theoretical proof in support of Boltzmann-plank equation.	05
2	188	Obtain an equation for most probable distribution energy.	05
2	189	Explain Boltzmann distribution law.	05
2	190	Derive an equation for partition function by using the concept of Boltzmann-distribution law.	05
2	191	Determine the thermodynamic parameters internal energy and entropy from partition function.	05
2	192	Determine the thermodynamic parameters Helmholtz free energy, pressure and Gibbs free energy from partition function.	05
2	193	Determine the thermodynamic parameters enthalpy and Gibbs free energy from partition function.	05
2	194	How to determine the partition function by using the concept of Boltzmann-distribution function.	05
2	195	Obtain an equation for translational partition function for monoatomic and polyatomic gaseous	05
2	196	Obtain an equation for rotational partition function for monoatomic and polyatomic gaseous molecule	05
2	197	Obtain an expression for vibrational partition function for monoatomic and polyatomic gaseous	05

2	198	Obtain an expression for electronic partition function for monoatomic gaseous molecule	05
2	199	Draw phase diagrams for two component systems and explain briefly in which the two components form a stable compound with congruent melting point.	05
2	200	Draw phase diagrams for two component system and explain briefly in which the two component form compound with incongruent melting point	05
2	201	Explain the phase diagram for a system of three liquids.	05
2	202	What are primary and secondary batteries explain with examples.	05
2	203	Write a note on lithium batteries	05
2	204	What are storage batteries? Discuss the advantages and disadvantages of lithium storage batteries	05
2	205	What are batteries? Discuss their classification and characteristic features.	05
3	206	Write a note on mode of interaction of gamma radiation with matters.	05
3	207	Explain Photoelectric effect after interaction between X-rays and matter	05
3	208	Explain the dominant interaction of photons between 0.1 to 10 Mev.	05
3	209	Describe the interaction of gamma radiation with matter using Compton effect.	05
3	210	What is Compton wavelength? Write its equation, also, Explain the principle of pairs production in interacting of gamma radiation with matter.	05
3	211	Write a note on (i) Pair production (ii) Compton scattering	05
3	212	Describe the principle of gamma radiation interaction with matter in case of (i) Photoelectric effect (ii) Compton effect	05
3	213	Explain briefly units of radiation energy.	05
3	214	Write a note on (i) G value (ii) Units of radiation energy	05
3	215	Write a note on Fricke dosimeter.	05
3	216	Describe the preparation and procedure to reduce Ce^{+4} to Ce^{+3} ion using ceric sulphate dosimeter What is radiolysis? Also, explain what happens if radiolysis occurs in amino	05

		acid with an example.	
3	217	Explain how thiol group of amino acid becomes self protective during radiolysis.	05
3	218	Write the mechanism involved in radiolysis of cysteine	05
3	219	Summarize a report on radiolysis of biphenyl.	05
3	220	Illustrate elucidation of structural determination using radioisotope tracing technique	05
3	221	Summarize a report on determination of reaction mechanism using radioisotope as tracer.	05
3	222	Explain radioisotope tracer used in physicochemical problem of sparingly soluble substance.	05
3	223	Describe the role of C-14 isotope in the age determination.	05
3	224	Write a note on use of radioisotopes in medical field.	05
3	225	Summaries a report on hazards in radiochemical work.	05
3	226	Explain the precautions taken in radiochemical work.	05
3	227	Explain the classification of polymer with at least two examples.	05
3	228	Describe the linear, branched and network polymer?	05
3	229	Write a note on different type of polymerization?	05
3	230	Illustrate the difference between addition and condensation polymerization with an example.	05
3	231	Distinguish between free radical and Ionic polymerization?	05
3	232	Summarize coordination polymerization and ring opening polymerization?	05
3	233	Write a note on polymerization and its classification with an example for each?	05
3	234	Explain molecular weight and size of polymers with an example.	05
3	235	Describe the concept of average molecular weight of polymer Explain determination of molecular weight of polymer by osmotic pressure method.	05

3	236	Explain the concept like number, weight and viscosity of average molecular weight of polymer.	05
3	237	Illustrate the determination of molecular weight of polymer by viscosity method.	05
3	238	Describe the determination of molecular weight of polymer by sedimentation method.	05
3	239	Write a note on determination of molecular weight of polymer by light scattering method.	05
3	240	An equal number of protein mixture containing Hemoglobin 15.5 Kg/mol, Ribonuclease 13.7 Kg/mol, Myoglobin 17.2 Kg/mol.	05
3	241	Calculate number average molar mass (M_n) and weight average molar mass (M_w) of protein solution.	05
3	242	Make the difference between gel electrophoresis and zone electrophoresis	05
3	243	Write a note on zone electrophoresis	05
3	244	Explain the process of gel electrophoresis	05
3	245	Discuss the application of electrophoresis in qualitative and quantitative study of protein	05
3	246	Describe the principle of electrophoresis and zone electrophoresis in detail.	05
3	247	Illustrate the application of zone electrophoresis in qualitative and quantitative study of protein.	05
3	248	Describe the process of (i) Zone electrophoresis (ii) Gel electrophoresis	05
3	249	Summarize the difference between electro osmosis and streaming potential.	05
3	250	Explain the determination of isoelectric point of a protein.	05
3	251	Write the biological significance of electro osmosis	05
3	252	Explain the process of streaming potential and hence write its biological significance	05
3	253	Write a note on electro osmosis and its biological significance	05
3	254	Discuss the biological significance of Donnan membrane phenomenon	05

3	255	What is donnan membrane phenomenon? And hence explain its biological application	05
3	256	Write a note on (i) Donnan membrane phenomenon. (ii) Streaming potential	05
3	257	Describe the involvement of micelles during digestion and absorption of lipids.	05
3	258	Summarize the diffusion of solute of across bio membrane.	05
3	259	Analyze the application of diffusion of solute across bio membrane.	05
3	260	Write a note on (i) Role of micelles in digestion of lipids (ii) Diffusion of solute	05
3	261	Make the difference between salting in and salting out of protein	05
3	262	Illustrate the application of osmo-regulation in evolution of excretory system of organisms.	05
3	263	Describe osmotic behavior of cell and its application in the evolution of excretory system of organisms	05
3	264	Write a note on salting in and salting out of protein.	05
3	265	Write the reducible representation of water and ammonia molecule.	05
3	266	Discuss the character table for C_{2h} using GOT and identify its Mulliken symbol.	05
3	267	A polymer contains 1,2,3,4 and 5×105 respectively.Calculate number of average weight, average molecular weight and polydispersity index.	05
3	268	How isotropic traces are used for structure determination	05
3	269	Write the symmetry elements for benzene, pyridine, Boric acid and eclipsed ferrocene.	05

Question Paper Pattern- Model Question Paper

Q.P Code: 50442

St. Philomena's College (Autonomous) Mysore

IV Semester M.Sc. - Final Examination : May - 2019

Subject: CHEMISTRY : PHYSICAL CHEMISTRY

Title: Symmetry, Group theory, Statistical, Phase rule and Radiation Chemistry and Electrophoresis

Time: 3 Hours **Max Marks: 70**

Instruction to the Candidates: *Answer PART - A and any FIVE questions from PART - B*

PART –A

Answer the following: **10×2=20**

1. a. Prove that numbers ± 1 and $\pm i$ form a group under multiplication.
   b. Give the principle of zone electrophoresis.
   c. Identify the point group of following molecules and write all symmetry elements:
      i) NH_3 ii) H_2O
   d. What is co-ordination polymerization? Give an example.
   e. Define partition function of a system.
   f. State Gibbs – phase rule and mention any one of its applications.
   g. Define the units, Sievert and Rad.
   h. Write the classification of batteries.
   i. P:T $C_2 \cdot \sigma_n = i = S_2$ by considering transdichloroethylene.
   j. Write the matrix representation for C_2, σ_{yz}, i and E symmetry elements.

PART –B

Answer any FIVE of the following: **5×10=50**

2. a. Discuss the character table for C_{2h} using goT and identify its Mulliken symbol. **05**
   b. A polymer contains 1, 2, 3 and 4 molecules having molecular weights 10^5, 2×10^5, 3×10^5 and 5×10^5 respectively. Calculate number of average weight, average molecular weight and polydispersity index. **05**

PTO

3. a. Deduce an expression for the relation between entropy and thermodynamic probability. **05**

   b. Discuss IR activity of H_2O molecule by its character table. **05**

4. a. Explain the applications of phase rule to two-component system with incongruent melting point. **05**

   b. What are lithium batteries? How do you calculate ΔG and ΔH using energetics of cell reactions? **05**

5. a. How isotropic traces are used for structure determination. **05**

   b. What is gel-electrophoresis? Outline the experimental determination of the isoelectric point of glycine. **05**

6. a. S.T character system of C_{3v} obey rules of irreducible representation. **05**

   b. Evaluate the translational partition function for a manoatomic and polyatomic gaseous molecules **05**

7. a. Discuss the determination of molecular weight by light scattering method. **05**

   b. Calculate rotational partition function of H_2 at $300^\circ K$? **05**

   (Given: $\sigma = 2$ *for* H_2) $\overline{B}$ for $H_2 = 60.864/cm$.

8. a. Write the Biological significance of donnan membrane phenomenon. **04**

   b. Write the symmetry elements for **06**

   i) 

   ii) H_3BO_3

   iii) Eclipsed ferrocene

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